

Machine Learning & AI Guide

A compact knowledge base for the Agentic RAG Document Assistant project.

1. What is Machine Learning?

Machine Learning (ML) is a branch of artificial intelligence in which computers learn patterns from data and use those patterns to make predictions or decisions. A typical ML workflow includes collecting data, cleaning data, selecting features, training a model, evaluating it, and using the trained model for prediction.

2. Supervised Learning

Supervised learning uses labeled data. Each training example contains input features and a known target. Classification predicts categories such as spam or not spam. Regression predicts numerical values such as house prices or sales.

3. Linear Regression

Linear Regression models the relationship between input variables and a continuous target using a linear equation. It is commonly used for problems such as predicting prices, revenue, or demand. Mean Squared Error (MSE) and Mean Absolute Error (MAE) are common regression metrics.

4. Logistic Regression

Logistic Regression is mainly used for classification. It estimates the probability of a class and commonly uses the sigmoid function for binary classification. Accuracy, precision, recall, F1-score, and ROC-AUC can be used to evaluate classification models.

5. Decision Trees

A Decision Tree makes predictions by repeatedly splitting data according to feature conditions. It is easy to interpret and can handle numerical and categorical features. Very deep trees can memorize the training data and overfit.

6. Random Forest

Random Forest is an ensemble method that combines many decision trees. Each tree is trained using randomized data and feature selection. Combining many trees generally makes the model more robust than a single decision tree.

7. Overfitting and Underfitting

Overfitting occurs when a model learns the training data too closely and performs poorly on unseen data. Underfitting occurs when a model is too simple to capture important patterns. Regularization, cross-validation, simpler models, more training data, and appropriate feature selection can help reduce overfitting.

8. Train, Validation, and Test Data

Training data is used to learn model parameters. Validation data can be used to tune hyperparameters and compare approaches. Test data is kept separate and is used to estimate performance on unseen data. Data leakage should be avoided because it can produce unrealistically good evaluation results.

9. Cross-Validation

Cross-validation repeatedly divides the available training data into training and validation portions. In k-fold cross-validation, the data is divided into k folds and each fold is used as validation once. This gives a more reliable estimate of model performance than relying on one split.

10. Neural Networks

A neural network consists of layers of connected neurons. The input layer receives features, hidden layers learn representations, and the output layer produces predictions. Weights are adjusted during training using backpropagation and an optimization algorithm.

11. Activation Functions

Activation functions introduce non-linearity into neural networks. ReLU is widely used in hidden layers because it is simple and computationally efficient. Sigmoid is often used for binary output probabilities, while Softmax is commonly used for multi-class classification.

12. CNN

A Convolutional Neural Network (CNN) is especially useful for image and spatial data. Convolutional layers learn local patterns such as edges and textures, while pooling layers can reduce spatial dimensions. CNNs are widely used in image classification and object recognition.

13. RNN and LSTM

Recurrent Neural Networks (RNNs) are designed for sequential data. LSTM networks are a type of RNN designed to better preserve information over longer sequences. They have been used for text processing, time-series forecasting, and sequence classification.

14. Natural Language Processing

Natural Language Processing (NLP) enables computers to process human language. Common tasks include sentiment analysis, text classification, summarization, translation, and question answering. Text can be represented using techniques such as TF-IDF, word embeddings, and transformer-based embeddings.

15. Embeddings

An embedding represents text or another object as a numerical vector. Similar meanings tend to have vectors that are close together in embedding space. Embeddings are important in semantic search and Retrieval-Augmented Generation because they allow a system to find relevant content based on meaning rather than exact keywords.

16. Large Language Models (LLMs)

A Large Language Model is a neural network trained on large amounts of text to understand and generate language. LLMs can answer questions, summarize information, generate code, and perform many language tasks. An LLM can also be connected to external tools and retrieved information instead of relying only on its internal knowledge.

17. RAG

Retrieval-Augmented Generation (RAG) combines information retrieval with an LLM. Documents are split into chunks, converted into embeddings, and stored in a vector database. When a user asks a question, relevant chunks are retrieved and supplied to the LLM as context before it generates an answer.

18. Vector Databases

A vector database stores numerical embeddings and supports similarity search. FAISS is a popular library for efficient similarity search over vectors. In a RAG application, the vector store helps retrieve document chunks that are semantically related to a user's question.

19. Agents and Agentic AI

An AI agent is a system in which an LLM can decide what action to take to complete a task. An agent may choose between tools such as document search, a calculator, or another API. Agentic RAG can allow an agent to decide whether retrieval is necessary and can perform another retrieval step when the first result is insufficient.

20. Agentic RAG Example

Suppose a user asks, 'What causes overfitting and how can it be reduced?' The agent can decide that the document collection should be searched. The retriever finds relevant chunks about overfitting and regularization. The LLM then uses those chunks to generate an answer and can provide the document name as a source.

21. Important RAG Concepts

Chunk size controls how much text is stored in each retrieval unit. Chunk overlap preserves context between neighboring chunks. Top-k retrieval controls how many relevant chunks are returned. Embedding quality affects semantic search quality. A good RAG system should retrieve relevant information while minimizing irrelevant context.

22. Key Interview Questions

What is RAG? Why are embeddings used? What is a vector database? What is the difference between keyword search and semantic search? What is an LLM? What is an AI agent? How does an agent decide which tool to use? What is chunking? Why is chunk overlap useful? How can RAG reduce hallucinations? What happens if the retriever returns irrelevant documents?